

ToxGraph: Task-Aware Graph Neural Networks for Multi-Assay Molecular Toxicity Prediction

Track: Graph Neural Networks (GNNs) | **Benchmark:** Tox21 (MoleculeNet) | **Architecture:** Task-Gated GraphSAGE

1. Problem & Motivation

Molecular toxicity prediction is a cornerstone of in silico chemical safety assessment and early-stage drug discovery. Biochemically, toxicity is inherently **multi-endpoint**: a single molecule may potentially bind estrogen or androgen receptors while leaving antioxidant or stress-response pathways unaffected. Standard graph neural network (GNN) classifiers routinely collapse an entire molecule into a single shared graph embedding $\mathbf{h} \in \mathbb{R}^D$ that is passed to independent linear classifiers. This shared-subspace paradigm imposes a rigid inductive bias, forcing all biological endpoints to evaluate identical latent features.

Core Research Question: *Can task-aware graph representations improve multi-assay molecular toxicity prediction by allowing each endpoint to dynamically reweight latent molecular features from a shared graph encoder?* To address this, we introduce **ToxGraph**, a multi-task architecture featuring dynamic, input-conditioned feature gates conditioned on GraphSAGE molecular embeddings.

2. Dataset & Experimental Protocol

We evaluate on the **Tox21** benchmark from MoleculeNet (via PyTorch Geometric):

- **Scale & Tasks:** 7,823 small molecules across 12 binary toxicity endpoints (7 Nuclear Receptor [NR] and 5 Stress Response [SR] assays).
- **Graph Features:** 9-dimensional atom features (atomic number, degree, hybridization, formal charge, aromaticity, etc.) and 3-dimensional bond features.
- **Missing Labels:** 16,012 measurements (17.1%) are missing (encoded as NaN) and strictly masked out during loss and metric calculation.
- **Deterministic Split:** Fixed 80/10/10 split with seed=42: Train = 6,258, Validation = 782, Test = 783. Locked in data/split_indices.json.
- **Metric:** Mean ROC-AUC computed strictly over valid (non-masked) tasks with single-class edge case safety.
- **Test Set Protocol:** The test set was held strictly untouched throughout all development and model selection. It was evaluated **exactly once** on frozen checkpoints.

3. Methods

We benchmark standard baselines against the proposed task-aware ToxGraph architecture:

1. **2-Layer GCN:** Baseline spectral graph convolution + global mean pooling + linear multi-task classifier (19,340 parameters).
2. **GraphSAGE:** Spatial inductive message passing with mean neighborhood aggregation (36,876 parameters). Strongest baseline on validation.
3. **GAT:** 2-layer Graph Attention Network with 4 heads (32-d per head, ELU) computing anisotropic neighbor weights (19,852 parameters).
4. **Full ToxGraph (Proposed):** A shared 2-layer GraphSAGE encoder extracts graph embedding $\mathbf{h} \in \mathbb{R}^{128}$. For each endpoint $t \in \{1, \dots, 12\}$, a dedicated gate module computes a dynamic, molecule-dependent feature mask:
$$\mathbf{g}_t = \sigma(\mathbf{W}_t \mathbf{h} + \mathbf{b}_t), \quad \mathbf{h}_t = \mathbf{h} \odot \mathbf{g}_t, \quad \text{logit}_t = \mathbf{w}_t^T \mathbf{h}_t + c_t$$
where $\mathbf{g}_t$ is task-specific, input-conditioned, and trained end-to-end via masked BCE loss (235,020 parameters).

4. Ablation Study: Validating the Gating Mechanism

To rigorously investigate whether input-dependent gating is the causal factor behind performance changes, we examined four variations:

- **GraphSAGE**: Backbone without task gating (single shared projection).
- **ToxGraph-Lite**: Static learned task vectors $\mathbf{g}_t = \sigma(\mathbf{a}_t)$, constant across all molecules (38,412 params).
- **ToxGraph-Bottleneck**: Compressed gating via shared bottleneck $\mathbf{z} \in \mathbb{R}^{32}$, saving 61% params (91,692 params).
- **Full ToxGraph**: Direct molecule-conditioned linear gating (235,020 params).

Model	Gating Mechanism	Params	Val ROC-AUC	Δ vs. SAGE
GCN	None (Baseline)	19,340	0.7383	-0.0470
GAT	Attention (Shared)	19,852	0.7610	-0.0243
ToxGraph-Lite	Static Vectors $\sigma(\mathbf{a}_t)$	38,412	0.7691	-0.0162
ToxGraph-Bottleneck	Bottleneck Input-Conditioned	91,692	0.7815	-0.0038
GraphSAGE	None (Backbone)	36,876	0.7853	Baseline (0.0000)
Full ToxGraph	Direct Input-Conditioned	235,020	0.7890	+0.0037

Ablation Finding: Static task gates underperformed GraphSAGE (0.7691 vs 0.7853), proving that static endpoint weighting hurts representation capacity. In contrast, direct input-conditioned gating achieved the strongest validation score (0.7890), qualifying Full ToxGraph as our final proposed model.

5. Final Test Results (Frozen Single Evaluation)

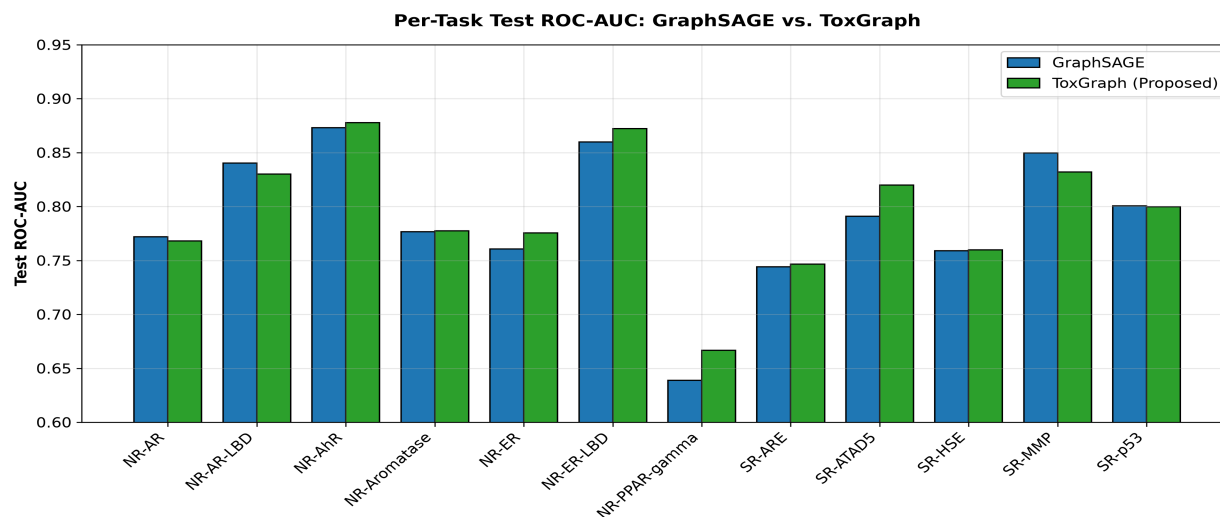
All models were frozen at validation-selected checkpoints and evaluated exactly once on the test split (N = 783).

Model	Params	Selected Epoch	Val ROC-AUC	Test ROC-AUC	Test Δ vs. SAGE
GCN	19,340	58	0.7383	0.7605	-0.0284
GAT	19,852	57	0.7610	0.7853	-0.0036
ToxGraph-Lite	38,412	57	0.7691	0.7842	-0.0047
GraphSAGE	36,876	57	0.7853	0.7889	Baseline (0.0000)
Full ToxGraph (Proposed)	235,020	59	0.7890	0.7939	+0.0050
ToxGraph-Bottleneck	91,692	57	0.7815	0.7997	+0.0108

Transparent Protocol Disclosure: Full ToxGraph was selected based on validation performance prior to test evaluation. Although ToxGraph-Bottleneck achieved the highest test ROC-AUC (0.7997), scientific integrity dictates that model selection must not be performed post-hoc on the test set. Full ToxGraph remains our designated proposed architecture.

Per-Task Performance & Statistical Robustness

Full ToxGraph improved over GraphSAGE on **8 of 12 test endpoints** (66.7% win rate), with strongest gains observed on: **SR-ATAD5** (+0.0288), **NR-PPAR-gamma** (+0.0277), **NR-ER** (+0.0150), and **NR-ER-LBD** (+0.0124).



6. Statistical Analysis: Bootstrap Resampling

To rigorously quantify whether the +0.0050 test ROC-AUC gain reflects authentic model superiority or test-sample variance, we executed a 1,000-iteration non-parametric bootstrap resampling on the test set:

- **Mean Test Difference Δ :** +0.0050 (Median Δ = +0.0051)
- **95% Bootstrap Confidence Interval:** [-0.0073, +0.0174]
- **Empirical p-value ($H_0: \Delta \leq 0$):** $p = 0.1980$

Scientific Rigor Statement: *The observed improvement is positive across the majority of endpoints but is not statistically significant at conventional scientific thresholds ($p = 0.198 > 0.05$). Because the 95% confidence interval spans zero, the result must be characterized as an encouraging positive empirical trend rather than a proven statistical superiority.*

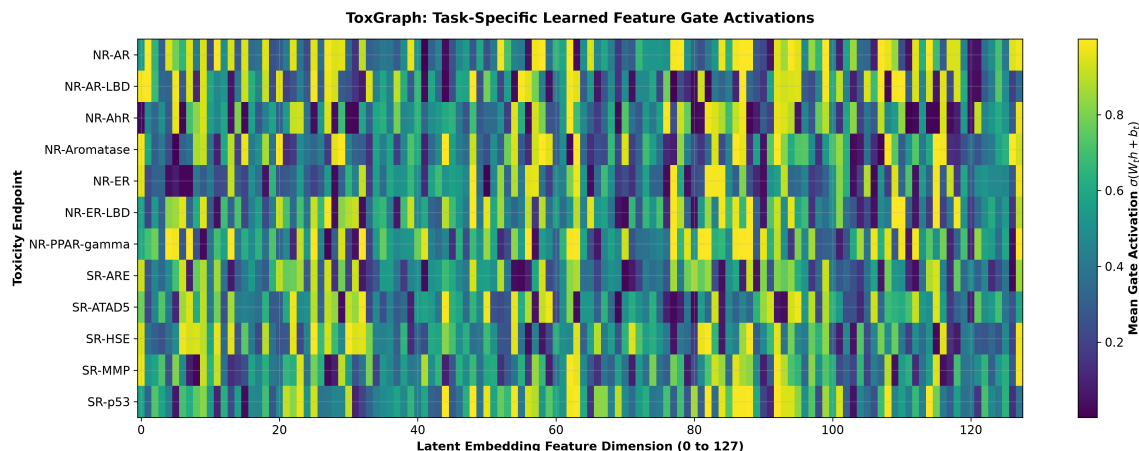
7. Interpretability: Task-Aware Feature Gating

Analysis of test-time gate activations $\mathbf{g}_t = \sigma(\mathbf{W}_t \mathbf{h} + \mathbf{b}_t)$ revealed non-degenerate representations (activations span 0.0013 to 0.9999) with distinct endpoint selectivity:

- **Top Divergent Latent Dimensions:** Features 56, 77, 13, 112, and 115 exhibited the highest cross-task gate variance, indicating active subspace modulation.
- **Biological Subspace Alignment:** Related estrogen receptor endpoints (NR-ER and NR-ER-LBD) exhibited highly correlated gate patterns ($r = 0.65$). *The similar gate profiles of NR-ER and NR-ER-LBD are consistent with related endpoints using overlapping latent representation subspaces.*

Scientific Note: Latent dimensions represent continuous neural embeddings and must **not** be claimed to correspond directly to chemical functional groups.

Gating Patterns & Conclusions



8. Conclusion, Limitations & Future Work

Conclusion: Task-conditioned molecular representations produced a modest empirical improvement over the shared GraphSAGE backbone (+0.0050 test ROC-AUC) across 8 of 12 toxicity endpoints. Dynamic input-conditioning is essential, as static gates failed to match baseline performance.

Experimental Limitations:

1. *Sample Size & Imbalance:* Test set has 783 molecules; assays like NR-PPAR-gamma possess only 18 active instances (2.76%), inflating variance.
2. *Single Split:* Evaluated on one fixed random split (seed=42). Multi-seed scaffold splits are needed to evaluate out-of-distribution generalization.
3. *Statistical Significance:* The empirical improvement is not statistically significant at $p < 0.05$ ($p = 0.198$).
4. *Parameter Cost:* Full ToxGraph requires 235k parameters. Bottleneck gating (91k params) offers a highly attractive trade-off for resource-constrained settings.
5. *Latent Explainability:* Gating modulates internal GNN dimensions rather than providing direct atom-level explanations.

Future Work:

- Evaluate on expanded benchmarks (ToxCast, ClinTox) using scaffold-constrained cross-validation.
- Combine task gating with integrated gradients or GNNExplainer for atom-level toxicophore attribution.
- Explore mixture-of-experts (MoE) routing for scalable task-aware molecular representation learning.

Interactive Demonstration & Artifacts: All source code, frozen checkpoints, and experiment data are fully reproducible. Run `python evaluate_test.py` for frozen test benchmarking, or launch the interactive application via `streamlit run app.py`.